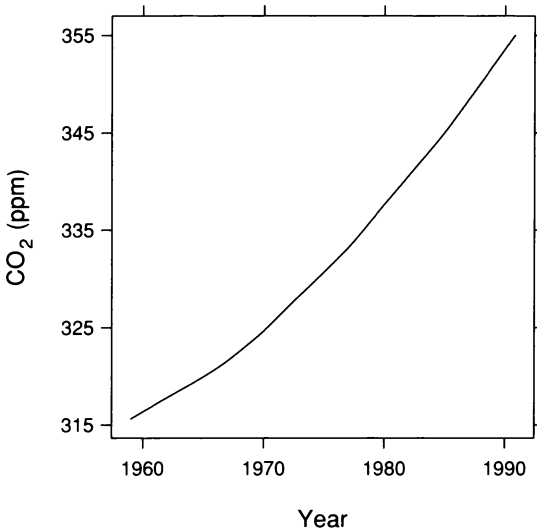
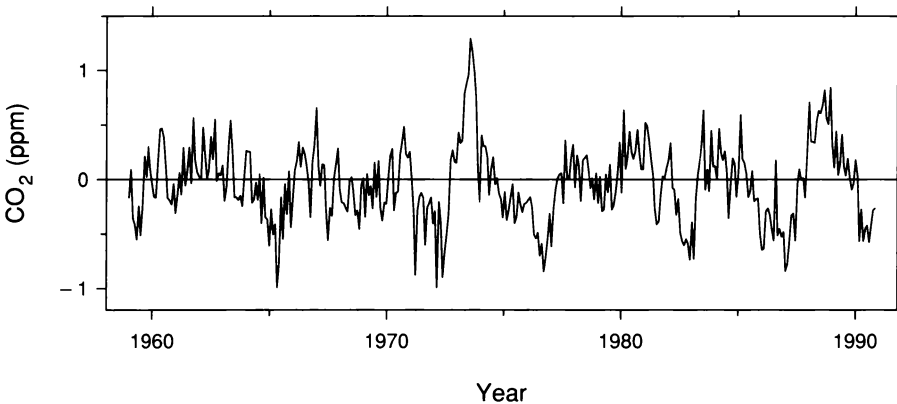


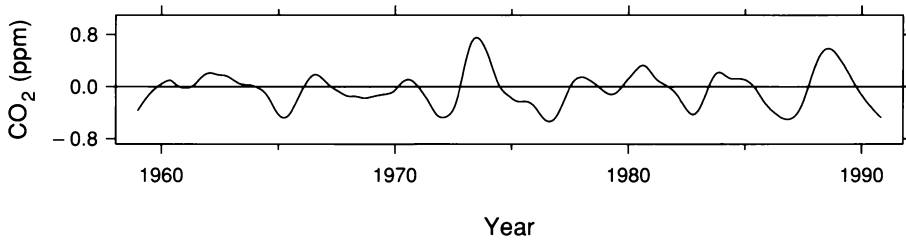
Figure 3.77 graphs a loess trend fit to the residuals with $\ell = 101$ and $\lambda = 1$. Figure 3.78 graphs the residuals from the trend fit. There are oscillations in the data — a rising and falling of the general level with periods of about 2 to 5 years. Figure 3.79 shows a loess fit to these residuals with $\ell = 35$ and $\lambda = 2$. Since we have fitted two time components to the residuals from the seasonal component, several passes of repeated loess fitting are in order. But as with the melanoma data, the final fits are very close to the initial.



3.77 The display graphs a loess trend fit to the residuals from the seasonal component. The local segments are banked to 45° .



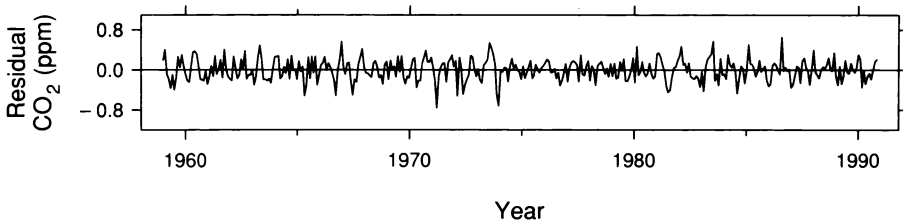
3.78 The residuals from the loess trend fit are displayed. The aspect ratio is 1/3 to enhance the perception of the oscillations in the data.



3.79 The curve is an oscillatory component fitted to the residuals from the trend fit. The local segments are banked to 45° .

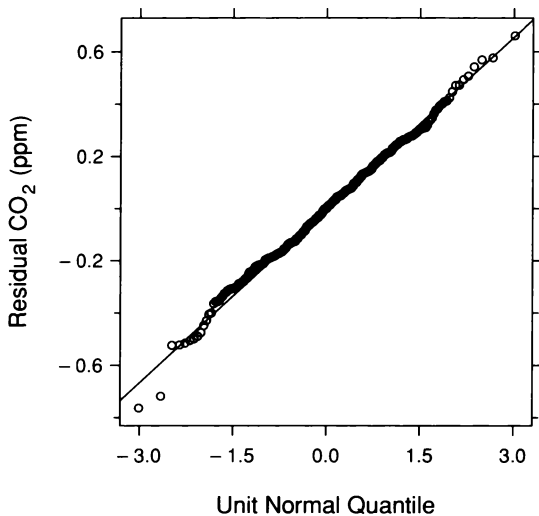
The trend component in Figure 3.77 is convex; the rate of increase of CO_2 is increasing. The oscillatory component in Figure 3.79 has about 9 complete cycles over 32 years, so the average period is about 3.5 years. This variation is associated with changes in the southern oscillation index, a measure of the difference in atmospheric pressure between Easter Island in the South Pacific and Darwin, Australia. Changes in the index are also associated with changes in climate. For example, when the index drops sharply, the trade winds are reduced and the temperature of the equatorial Pacific increases. Many regions, including South America, can be drastically affected. In years when such an event occurs, the ocean warming normally begins in December, so the event has become known as El Niño, the child [59].

The final residuals from the three components fitted to the CO₂ data — trend, 3.5-year, and seasonal — are graphed in Figure 3.80. No striking time effect remains.



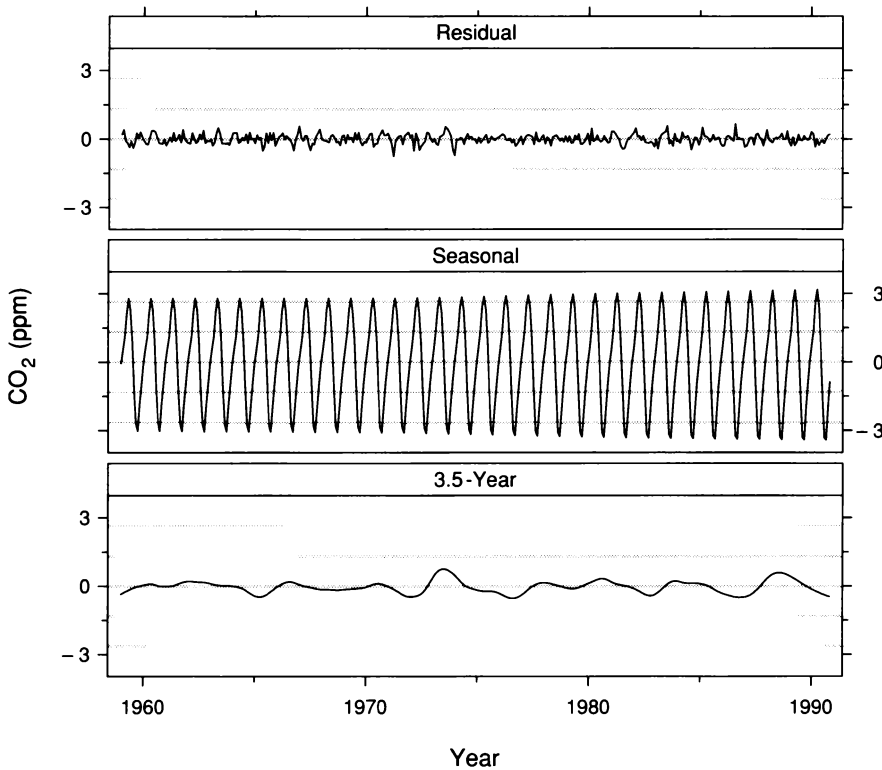
3.80 The residuals from the trend, 3.5-year, and seasonal components are displayed. The aspect ratio is 0.1. Banking local segments to 45° results in an aspect ratio that is too small.

Figure 3.81, a normal q-q plot, shows that the distribution of the residuals is well approximated by the normal. Thus we need not fear that outliers or more pervasive leptokurtosis is damaging our fits.



3.81 The CO₂ residuals are displayed by a normal q-q plot.

Finally, Figure 3.82 graphs the 3.5-year, seasonal, and residual components by an r-f spread plot. The trend component has been omitted since, ominously, its variation is so much greater than that of the other three. Its overall variation is 39.4 ppm, while the largest overall variation of the other three, that of the seasonal, is only 6.6 ppm.



3.82 An r-f spread plot compares the variation in the 3.5-year component, the seasonal component, and the residuals.

For the Record: The Details of Seasonal Loess

In seasonal loess, the long-period component is fitted to prevent variation with periods greater than that of the seasonal from leaking into the seasonal. The long-period component is not thought of as an output of the procedure; its purpose is just to facilitate the production of the seasonal component.

Let the time series be Y_t for times $t = 1$ to n . Suppose the number of observations in each period, or cycle, of the seasonal component is p . For example, if the series is monthly with a yearly periodicity, then $p = 12$. Suppose also that the choices of the parameters for the seasonal fitting are $\ell = \ell_{(s)}$ and $\lambda = \lambda_{(s)}$. As we will see, all other loess parameters used in seasonal loess are chosen automatically.

As stated earlier, seasonal loess is an iterative procedure, with several passes. On each pass, the seasonal component and the long-period component are fitted. Suppose $S_{t,k}$ and $L_{t,k}$ are the seasonal and long-period components at the end of the k th pass.

The result of the first stage of pass $k + 1$ is an updated seasonal component. First, residuals, $Y_t - L_{t,k}$, are computed. (On the first pass, no long-period component is available, so no subtraction is carried out.) Each cycle subseries of the residual series is fitted by loess with $\ell = \ell_{(s)}$ and $\lambda = \lambda_{(s)}$. Each loess fit is evaluated at all time points of the subseries and at two additional points: one time step ahead and one time step behind. For example, suppose that the series is monthly, that $p = 12$, and that the January subseries ranges from January 1943 to January 1997. Then the loess fit for January is evaluated at all time points from January 1942 to January 1998. The collection of fitted values for all of the subseries is a series, $C_{t,k+1}$, consisting of $n + 2p$ values that range from $t = -p + 1$ to $n + p$. Next, $C_{t,k+1}$ is filtered by a moving average of length p , followed by another moving average of length p , followed by a moving average of length 3, followed by a loess fit with $\lambda = 1$ and ℓ equal to the smallest odd integer greater than or equal to p . The output, $F_{t,k+1}$, is defined at time positions $t = 1$ to n because the three moving averages cannot extend to the ends, so in all, p positions are lost at each end. The fitted seasonal component from this pass is $S_{t,k+1} = C_{t,k+1} - F_{t,k+1}$ for $t = 1$ to n . $F_{t,k+1}$ is subtracted to help prevent long-period variation from entering the seasonal component.

The result of the second stage of pass $k + 1$ is an updated long-period component. First, residuals, $Y_t - S_{t,k+1}$, are computed. Then the residuals are fitted by loess with $\lambda = 1$ and

$$\ell = \frac{1.5p}{1 - 1.5\ell_{(s)}^{-1}}$$

rounded to an integer. The fitted long-period component from this step $L_{t,k+1}$, is the loess fit evaluated at the time points $t = 1$ to n .

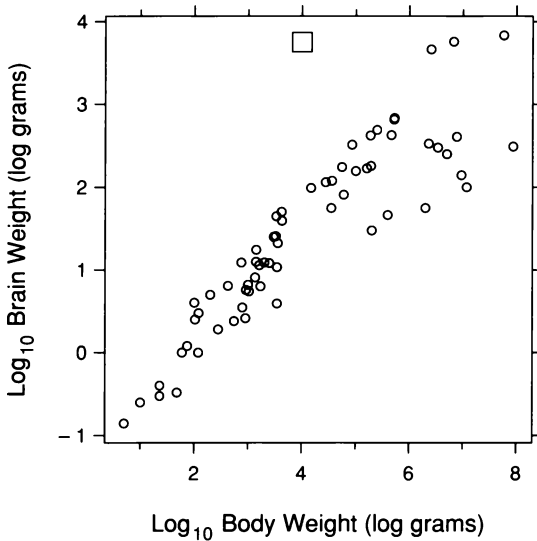
As with repeated loess fitting, bisquare can be used in seasonal loess by computing bisquare weights at the end of each pass and using them in the next pass. The weights after pass k are computed using the residuals, $Y_t - S_{t,k} - L_{t,k}$. Bisquare was not used for the CO₂ data because the residuals have a distribution that is well approximated by the normal.

In some cases, the change in the level of each subseries can be nearly flat through time but not quite constant. In other words, the seasonal component changes, but only very gradually. In such a case, it can make sense to take $\lambda_{(s)} = 0$, a value of λ not yet encountered in our account of loess fitting. This means a constant in time is fitted locally rather than a linear or quadratic function of time. While such a value of λ can make sense for time series applications, it almost never makes sense for the applications in which the factor is not time.

Other loess smoothing parameters in the above description of seasonal loess have been given values that work quite well in most applications. But aficionados of fitting seasonal components can experiment with these parameters as well.

3.13 Direct Manipulation: Labeling by Brushing

For some bivariate data sets, each observation has a label. Figure 3.83 is a scatterplot of log brain weight against log body weight for a collection of animal types. In this case, the label for each bivariate observation is an animal name.

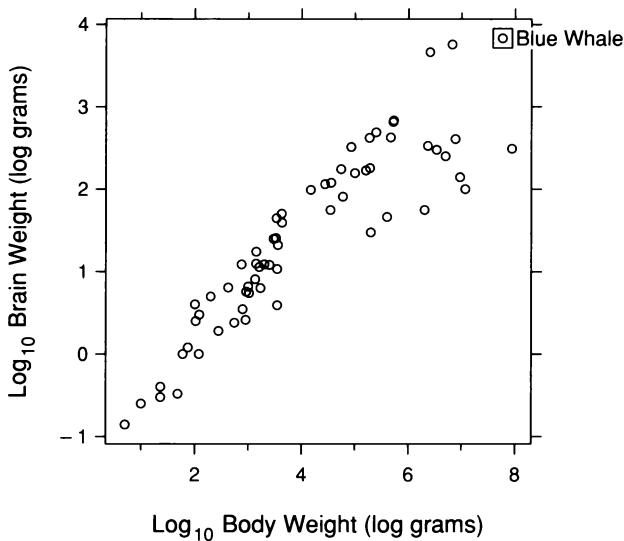


3.83 Log brain weight is graphed against log body weight for various animal types. The rectangle on the graph is the brush.

In the course of visualizing labeled data, it is often helpful to selectively label points identified as interesting from their location on a scatterplot. This typically cannot be achieved simply by labeling all points because the labels overlap and obscure one another. Because our selection of points to be labeled is based on visual identification, direct manipulation is ideal for labeling.

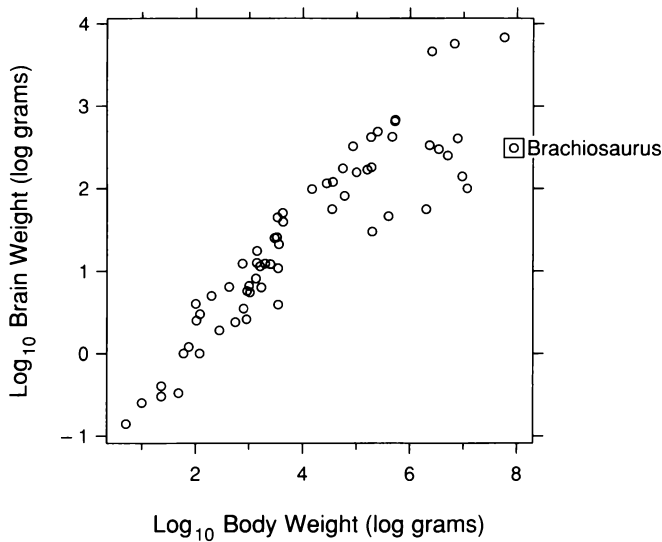
Brushing is a basic tool of direct manipulation that can be used for labeling and, as we shall see in later chapters, for other direct manipulation methods as well [6]. The rectangle at the top of the data region in Figure 3.83 is the *brush*, the central object in brushing. There are two important aspects of the control that the data analyst has of the brush. One is that it is moved around the screen by moving a mouse. The second is that it can be made any size or shape; in Figure 3.83, it has been made small and square.

The problem of labeling points on a scatterplot is solved quite nicely by brushing. When the brush covers one or more points, the label for each covered point appears. This is illustrated by Figure 3.84; we can now see that the animal with the largest brain weight is the blue whale.



3.84 Brushing with the labeling operation in the transient mode shows one label.

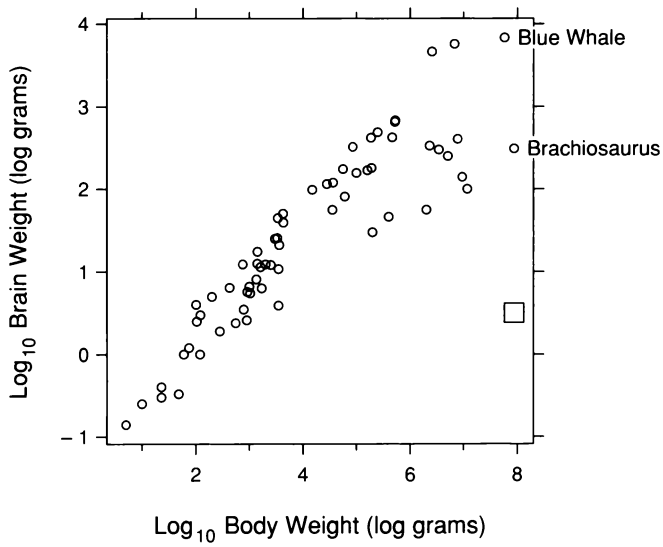
There are three modes in which the labeling operation can be carried out — *transient*, *lasting*, and *erase*. If the mode is *transient*, the label for a covered point disappears when the brush no longer covers it. This is illustrated by Figure 3.85. The brush has been moved from the blue-whale point to the point of an animal type with a slightly bigger body than the blue whale, but with a much smaller brain, clearly a dumb animal compared with the whale. As soon as the brush leaves the blue-whale point, the blue-whale label disappears. As soon as the brush touches the new point, the new label appears showing it is the genus *Brachiosaurus*.



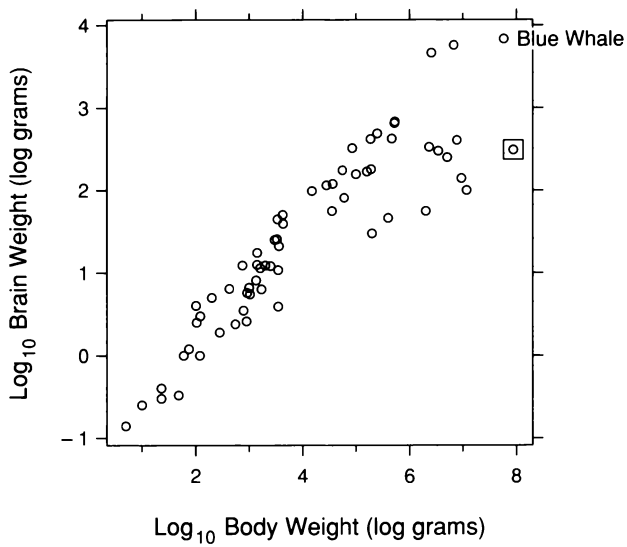
3.85 Brushing in the transient mode shows another label.

If the mode is *lasting*, the label remains even after the brush is removed from the point. This is illustrated by Figure 3.86. The brush has touched two points, and the labels remain even after the brush is moved from the points.

If the mode is *erase*, a lasting label is removed when the brush touches it. This is illustrated by Figure 3.87; the lasting *Brachiosaurus* label created in Figure 3.86 has been removed by covering the genus' point.



3.86 Brushing in the lasting mode has resulted in two lasting labels.



3.87 Brushing in the erase mode has removed one of the lasting labels.

3.14 Visualization and Probabilistic Inference

Rote Data Analysis

The experimenters who collected the ganglion data that were analyzed at the beginning of this chapter related CP ratio to area by fitting a least-squares line [60]. Probabilistic inference was carried out using a standard t-interval method that is based on an assumption that the deviations of the response from the line are a homogeneous random sample from a normal population distribution. Suppose this assumption is true. If we subtract the population slope from the slope estimate and then divide by a standard estimate of the population standard deviation of the slope estimate, the result has a t-distribution. This t-statistic can be used to form confidence intervals and to test the null hypothesis that the slope is zero. The experimenters quote the results of the hypothesis test in their account:

The linear relation between the C/P ratio and retinal area is highly significant (slope = 0.107 ± 0.010 ; $P < 0.001$) . . .

Such a quotation of a p-value is part of the ritual of science, a sprinkling of the holy waters in an effort to sanctify the data analysis and turn consumers of the results into true believers.

The rote data analysis of the experimenters has produced nonsense. The visualizations earlier in the chapter showed that a line does not fit the data. That ends matters right there. The standard t-interval method is not valid. The sampling variability that it prescribes for the estimate is not the actual variability. Once the lack of fit is appreciated, it is possible to make correct statements about variability, but in this example it makes no sense to characterize the probabilistic properties of a function estimate that does not fit the data.

The visualizations of the chapter also showed that a quadratic polynomial fits CP ratio without lack of fit, but there is monotone spread. So even with the right function, the t-interval method is inapplicable because the residuals are not homogeneous.

Finally, the visualizations showed that by taking the logarithm of CP ratio, the underlying pattern is linear, the spread no longer depends on location, and the distribution of the residuals is well approximated by the normal distribution. For this analysis, we can venture the standard methods to make probabilistic inferences.

Combining Prior Information and Data

Visualization is an effective framework for drawing inferences from data because its revelation of the structure of data can be readily combined with prior knowledge to draw conclusions. By contrast, because of the formalism of probabilistic methods, it is typically impossible to incorporate into them the full body of prior information.

Even when we have checked all assumptions that underlie a probabilistic inference, we must be prepared to ignore its results if knowledge not accounted for in the inferences appears to provide a better explanation of the behavior of the data. The fly data analyzed in Section 3.8 are one example. The actual temperatures of the incubators in the fly experiment are the causal factor, not the nominal temperatures. Fortunately, the experimenters report much information about the actual temperatures, and they wisely encourage us to incorporate this knowledge into our conclusions from the experiment:

Errors in this study may arise from fluctuating temperatures. . . .
Nominal temperatures in accompanying tables and figures are to
be considered with these remarks in mind.

The visualization suggested a temperature anomaly at 23°C and 25°C — a deviation from the overall linear pattern. But the information that the actual temperatures varied by substantial amounts, particularly the large variation at 23°C, combined with the information from the visualization, makes it quite clear that the deviations in the actual temperatures could easily account for the anomaly at 23°C and 25°C. In other words, combining the results of the visualization with the knowledge of the actual temperature variation leads to a conclusion that the dependence of facet number on actual temperature is likely linear.

R. A. Fisher analyzed these data with probabilistic methods [38]. In an exceedingly uncharacteristic moment of poor insight, he used the nominal temperatures and ignored the information about actual temperature variation offered up by the experimenters. He tested the null hypothesis that the underlying pattern in facet number is linear in nominal temperature. To do so, he chose a method of analysis — the analysis of variance — that is based on an assumption of normal distributions with the same standard deviation. Our visualization shows that this is a reasonable assumption for the nominal temperature distributions. Fisher concluded that the null hypothesis of a linear dependence of facet number on temperature is rejected because the p-value of the test is small. He reports:

The deviations from linear regression are evidently larger than what would be expected, if the regression were really linear . . . There can therefore be no question of the statistical significance of the deviations from the straight line, although the latter accounts for the greater part of the variation.

It is certainly true that a small p-value would be expected if the underlying pattern were not linear. But a small p-value could also result from other causes. One of them is a fluctuation in actual temperature, the likely cause [9].